FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IV

JEE (Advanced)-2020

PAPER – 1

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

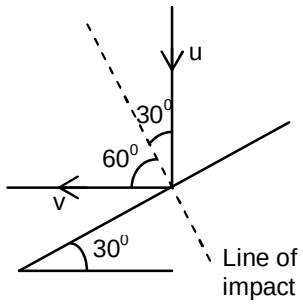
SECTION – A

1. B

Sol. Let velocity of ball just before collision is u and just after collision is v.

u sin 30° = v sin 60°

$$e = \frac{v\cos 60^0}{u\cos 30^0} = \frac{1}{3}$$



2. B

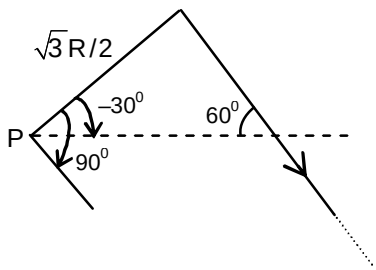
Sol. $\frac{1}{2}\rho v^2 + 0 + P_o = \left(P_o + \frac{mg}{A}\right) + \rho gh + \frac{1}{2}\rho O^2$

$$\therefore \quad v = \sqrt{2\left(gh + \frac{mg}{\rho A}\right)}$$

3. C

Sol. $B = \frac{\mu_o I}{4\pi\sqrt{3}\,R/2}\left[\sin 90^o + \sin(-30^o)\right]$

$$= \frac{\mu_o I}{4\sqrt{3}\,R}$$

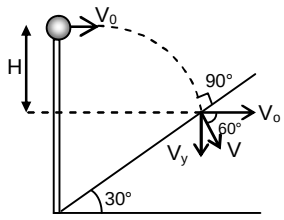


4. A

Sol. $v_y = \sqrt{2gH}$

$$\frac{v_y}{v_x} = \frac{\sqrt{2gH}}{v_o} = \tan 60^o = \sqrt{3}$$

$$v_0 = \sqrt{\frac{2gH}{3}}$$



5. BC

Sol. $v_L = v_C = v_R$;

$\Rightarrow x_L = x_C = R$

when inductor is short circuited

$Z = \sqrt{R^2 + x_C^2} = \sqrt{2}\,R$

$$\therefore I = \frac{30}{Z} = \frac{30}{\sqrt{2}R}$$

$$\therefore V_L = ix_L = \frac{30}{\sqrt{2}R} \times R = \frac{30}{\sqrt{2}}$$

$\therefore$ (A) is incorrect and with similar calculations (B) will be correct.

Here f_0 is the resonance frequency as $v_L = v_C$

$$\omega_0 = 2\pi f_0 = \frac{1}{\sqrt{LC}}$$

$$\frac{x_L}{x_C} = \frac{\omega L}{1/\omega C} = \omega^2 LC$$

Given $f = 2f_0$

$\Rightarrow \quad \omega = 2\omega_0$

$\therefore \quad \dfrac{x_L}{x_C} = 4$

$\therefore$ (C) is also correct.

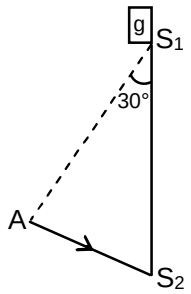
6. D

Sol. Extra phase change in glass = phae change in water of length AS_2.

$$\frac{2\pi t}{\lambda g} - \frac{2\pi}{\lambda\omega} t = \frac{2\pi}{\lambda\omega} \cdot \frac{2}{3} \sin 30^o$$

$$\Rightarrow t\left[\frac{3}{2} - \frac{4}{3}\right] = \frac{4}{3} \times \frac{1}{3}$$

$$\Rightarrow t = \frac{8}{3} \text{mm}.$$



7. ABC

Sol. $KE = h\nu - \phi$

8. C

Sol. Mass per unit length, $\mu = \dfrac{m}{\ell} = \dfrac{SA\ell}{\ell} = \rho A$

$\mu_s = \mu_{A\ell} = 78 \times 10^{-4}$ kg/m

$\therefore$ Speed of wave is same in both wire

$$V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{80\times10^4}{78}} = \frac{2\times10^2}{\sqrt{3.9}}$$

$$\nu_{min} = \frac{V}{\lambda_{max}} = \frac{200}{\sqrt{3.9}\times0.3} \quad \left[\frac{\lambda_{max}}{2} = 15 \text{ cm for C as a node}\right]$$

$$= 337.4 \text{ H}_2$$

9. AC

Sol. $$e = \frac{v_2 - v_1}{u_1 - u_2}$$

$$1 = \frac{v_2 - (-2)}{u_1}$$

$$u_1 = v_2 + 2$$

$$u_1 = 1(-2) + 5(u_1 - 2)$$

$$u_1 = -2 + 5u_1 - 10$$

$$u_1 = \frac{12}{4} = 3 \text{m/s}$$

$$v_2 = 1 \text{ m/s}$$

Kinetic energy of the centre of mass = $\frac{1}{2}\times(1+5)\times\left(\frac{3}{1+5}\right)^2 = \frac{3}{4}\text{J}$

10. AC

Sol. Velocity of insect w.r.t. strip = $\frac{\ell}{t}$

Let strip moves with speed ν

Initial momentum was 0

$$\Rightarrow \quad 0 = m\left(\frac{\ell}{t} + v\right) + Mv$$

$$v = -\frac{m\ell/t}{M+m} \text{ i.e. } \frac{m\ell/t}{M+m} \text{ toward left}$$

$\therefore$ Velocity of insect w.r.t. ground = $\frac{\ell}{t} - \frac{m\ell/t}{M+m} < \frac{\ell}{t}$.

11. BC

Sol. Cylinder will move downwards as mg sin θ > μ mg cos θ although frictional force acts upwards.

12. BCD

Sol. Apply Faraday law for direction of induced current.

SECTION – C

13. 00002.00

Sol. We have relation $M\frac{dv}{dt} = F_{ext} + v_{rel}\frac{dM}{dt}$.

In the question, we have the force by the gravity as the external force. As the rocket is to lift from the ground the minimum acceleration of the rocket to do so is, o, beyond which, with a slight increase it will zoom into the sky.

Therefore, $\frac{dv}{dt} = 0$

Putting these into the equation, we obtain

$$0 = Mg + v_{rel}\left(-\frac{dM}{dt}\right)$$

($\frac{dM}{dt}$ is rate of decrease of mass hence negative. Mg is in the direction of the relative velocity of the fuel with respect to the rocket)

Hence, $\frac{dM}{dt} = \frac{M}{V}g = \frac{400 \times 10}{2 \times 1(10^3)} = 2 \text{ kg/sec}$

14. 00008.00

Sol. Let u is the velocity of A and B along the string

$10 \cos 30 = 2u$

$\Rightarrow u = \frac{5\sqrt{3}}{2} m/s$

Let v is velocity of A $\perp$ to the string.

$10 \sin 30 = v$

$\Rightarrow v = 5$ m/s

$\therefore \quad V_A = \sqrt{u^2 + v^2} = \frac{\sqrt{175}}{2}$

$KE_A = \frac{175}{8} J$

15. 00003.75

Sol. $T = \frac{3}{8} M \omega^2 L^2$

$$\text{Stress} = \frac{F_\perp}{(A/\cos\theta)} = \frac{3}{8}\frac{M\omega^2 L^2 \cos^2\theta}{A}$$

16. 00002.17

Sol. $\frac{2.17 + 2.17 + 2.18}{3} = 2.1733 \ldots \approx 2.17$ mm upto 3 significant figure.

17. 00000.89

Sol. $2g - T = 2a$ …(i)

$TR = I\alpha$ …(ii)

$a = R\alpha$ …(iii)

From (ii) and (iii) $T = \frac{Ia}{R^2}$

$\therefore \quad 2g = a\left(2 + \frac{I}{R^2}\right)$

$\Rightarrow \quad a = \frac{2g}{\left(2 + \frac{I}{R^2}\right)} = \frac{2 \times 9.8}{2 + \frac{0.2}{0.01}} = \frac{9.8}{11} \approx 0.89 \text{ m/s}^2$

18. 00000.75

Sol. $(Fs)_{max} = 20$ N. a_{max} of 20 kg black = 1 m/s^2

$\therefore$ both black move together till 6 sec.

$\therefore \quad \frac{dv}{dt} = \frac{5t}{30} \quad \Rightarrow \quad \int dv = \frac{1}{6} \frac{t^2}{2} \quad ; \quad V = \frac{3^2}{12} = \frac{9}{12} = 0.75$ m/s

Chemistry

PART – II

SECTION – A

19. B

Sol. Each photon strikes out one electron form the metal surface. Intensity will increase the number of photons, so number of photoelectrons increase.

20. C

Sol. In(C), the complex ions have zero charge. So there is no attraction between the two.

21. B

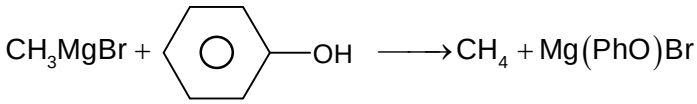
Sol. $\alpha = \frac{\Lambda_m}{\Lambda_m^0}$

22. B

Sol. $P_4O_{10} + 4HNO_3 \longrightarrow 2N_2O_5 + 4HPO_3$

23. ABCD

Sol.



24. AB

Sol. 2° and 3° amines are more basic than 1° amine.

25. ABD

Sol. It is derived from Le-Chateliers's principle.

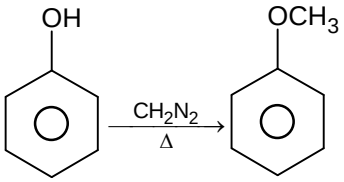
26. AB

Sol. $PH_4Cl \xrightarrow{\Delta} PH_3 + HCl$

$4H_3PO_3 \xrightarrow{\Delta} 3H_3PO_4 + PH_3$

27. ABD

Sol.



28. AB

Sol. $h = \sqrt{\frac{K_w}{K_a \times C}}$

H^+ remove OH^- forming weaker acid H_2O as compared to HCN.

29. AD

Sol. $\psi_{n,\ell,m}$ represents atomic orbitals.

30. AC

Sol. $t_{1/2} = \frac{a}{2k} \Rightarrow t_{1/2} \alpha \frac{1}{k}$

Pressure only changes the concentration of reactants. It has no effect.

SECTION – C

31. 01098.5

Sol. $P_{equilibrium} = p_B + 2p_C = 19.5$ atm

Since $p_B + 2p_C = 3x = 19.5$

or $x = 6.5$

$\therefore p_B = 6.5$ atm, $p_C = 13$ atm

$\therefore K_p = p_B \times (2p_C)^2 = 6.5 \times (2 \times 6.5)^2 = 1098.5$

32. 00122.5

Sol.



X is $CH_3 - CH(CH_3) - C(=O) - OH$

33. 00008.01

Sol. $\mu_{max} = q \times d = 1.602 \times 10^{-19} \times 5 \times 10^{-10} = 8.01 \times 10^{-29}$ cou.m

34. 00005.49

Sol. $pH = pK_a + \log \frac{[NaA]}{[HA]}$

or $6.49 = pK_a + \log \frac{1}{0.1} = pKa_a + 1$

$\therefore pK_a = 6.49 - 1 = 5.49$

35. 00154.5

Sol. X is mesitylene.

Y is Chloro mestiylene

36. 00036.5

Sol. $A = Na_2SO_4.10H_2O$, $B = Na_2SiO_3$, $C = SO_3$, $D = HCl$, $PPt = BaSO_4$

Mathematics

PART – III

SECTION – A

37. D

Sol. Let there be a value of k for which $x^3 - 3x + k = 0$ has two distinct roots between 0 and 1.

Let a, b be two distinct roots of $x^3 - 3x + k = 0$ lying between 0 and 1 such that $a < b$. Let $f(x) = x^3 - 3x + k$. Then $f(a) = f(b) = 0$. Since between any two roots of a polynomial f(x) there exist at least one root of its derivative $f'(x)$. Therefore $f'(x) = 3x^2 - 3$ has at least one root between a and b. But $f'(x) = 0$ has two roots equal to ± 1 which do not lie between a and b. Hence $f(x) = 0$ has no real roots lying between 0 and 1 for any value of k.

38. B

Sol. Total number of such function =27 out of which 12 are ONTO

39. C

Sol. Since coefficient of $x^2 + ax + 1$ are symmetrical so roots will be α, $\frac{1}{\alpha}$

$$\lim_{x\to 1/\alpha} \frac{\sin(x^2+ax+1)}{(\alpha x-1)} = \lim_{x\to 1/\alpha} \frac{\sin\left[\left(x-\frac{1}{\alpha}\right)(x-\alpha)\right]}{\left(x-\frac{1}{\alpha}\right)(x-\alpha)} \cdot \frac{\left(x-\frac{1}{\alpha}\right)(x-\alpha)}{(\alpha x-1)}$$

$$= \lim_{x\to 1/\alpha} \frac{(x-\alpha)}{\alpha} = \frac{\frac{1}{\alpha}-\alpha}{\alpha} = \frac{1-\alpha^2}{\alpha^2}$$

40. A

Sol. As, $f(x) = x^4\tan(x^3) - x\ln(1+x^2)$ is odd, $\frac{d^4f(x)}{dx^4} = 0$ at x = 0.

41. AD

Sol. Since $\int_a^b f(x)dx = (b-a)\int_0^1 f\{(b-a)x+a\}dx$,

$$\int_1^2 \sin x^2 dx = \int_0^1 \sin(x+1)^2 dx = \int_0^1 \sin(x^2+2x+1)dx$$

$$\int_{-4}^{4} \cos x^2\, dx = 8\int_0^1 \cos(8x-4)^2\, dx = 8\int_1^0 \cos 16(2x-1)^2\, dx.$$

42. ABCD

Sol. $A \subseteq A \cup B$

$\Rightarrow P(A) \le P(A \cup B) \Rightarrow P(A \cup B) \ge \frac{3}{4}$

Also, $P(A \cap B) = P(A) + P(B) - P(A \cup B)$

$\ge P(A) + P(B) - 1$

$= \frac{3}{4} + \frac{5}{8} - 1 = \frac{3}{8}$

Now,

$A \cap B \subseteq B$

$\Rightarrow P(A \cap B) \le P(B) = \frac{5}{8}$

$\therefore \quad \frac{3}{8} \le P(A \cap B) = \frac{5}{8}$

and

$P(A \cap B') = P(A) - P(A \cap B)$

$\Rightarrow \frac{3}{4} - \frac{5}{8} \le P(A \cap B') \le \frac{3}{4} - \frac{3}{8}$

$\Rightarrow \frac{1}{8} \le P(A \cap B') \le \frac{3}{8}$

$\therefore P(A \cap B) = P(B) = P(A' \cap B)$ [Using Equation (1)]

$\Rightarrow \frac{3}{8} \le P(B) - P(A' \cap B) \le \frac{5}{8}$

$\Rightarrow 0 \le P(A' \cap B) \le \frac{1}{4}$

43. AC

Sol. Given $a_1 = 2; \frac{a_n}{a_{n-1}} = \frac{a_{n-1}}{a_{n-2}}$

$\Rightarrow a_1, a_2, a_3, a_4, a_5, \ldots\ldots$ in G.P.

Let $a_2 = x$ then for n = 3 we have

$\frac{a_3}{a_2} = \frac{a_2}{a_1} = \frac{x_2}{2}$

$\Rightarrow a_1^2 = a_1 a_3$

$\Rightarrow a_3 = \frac{x^2}{2}$

i.e. $2, x, \frac{x^2}{2}, \frac{x^3}{4}, \frac{x^4}{8}, \ldots\ldots$ with common ratio $r = \frac{x}{2}$

given $\frac{x^4}{8} \le 162$

$\Rightarrow x^4 \le 1296 \le x \le 6$

Also $x \frac{x^4}{8}$ and are integers

$\Rightarrow x$ must be even then only $\frac{x^4}{8}$ will be an integer

Hence possible value of x is 4 and 6. ($x \neq 2$ as terms are distinct)

Hence possible value of $a_5 = \frac{x^4}{8}$ is $\frac{4^4}{8}, \frac{6^4}{8}$

44. AB

Sol. $3x - 6y + 2z + 5 = 0$

$-4x + 12y - 3z + 3 = 0$

Bisectors are $\frac{3x - 6y + 2z + 5}{\sqrt{9 + 36 + 4}} = \pm \frac{-4x + 12y - 3z + 3}{\sqrt{16 + 144 + 9}}$

The plane which bisects the angle between the planes that contains the origin is

$13(3x - 6y + 2z + 5) = 7(-4x + 12y - 3z + 3)$

$67x - 162y + 47z + 44 = 0$

Further, $3x(-4) + (-6)(12) + 2 \times (-3) < 0$

Hence, the origin lies in the acute angle.

45. AC

Sol. $\frac{a + b + c}{2\Delta} = \frac{2}{R} \Rightarrow R = 2r$ and $h_1 = \frac{2\Delta}{a} = 3r$

46. AD

Sol. Let $\hat{d}$ be the unit vector along the desired vector then $\hat{d}$ is along $\left(-2\hat{j} + 3\hat{k}\right) \times \left(-\hat{i} - 2\hat{k}\right)$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -2 & 3 \\ -1 & 0 & -2 \end{vmatrix} = 4\hat{i} - 3\hat{j} - 2\hat{k}$$

47. BD

Sol. $f'(x) = \frac{\sin x}{x}$

For $f'(x) = 0$, $\frac{\sin x}{x} = 0 \Rightarrow x = n\pi (n \in I, n \neq 0)$

$f''(x) = \frac{x\cos x - \sin x}{x^2}$

$f''(n\pi) = \frac{\cos n\pi}{n\pi} < 0$ if $n = 2k - 1$ and > 0 if $n = 2k, k \in I^+$

Hence, $f(x)$ has local maxima at $x = n\pi$, where n = 2k – 1 and

local minima at $x = n\pi$, n = 2k, where $k \in I^+$.

48. BCD

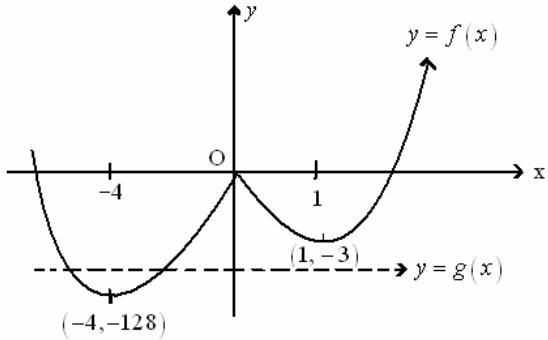
Sol. $x^4 + 4x^3 - 8x^2 = -k$

$\Rightarrow \quad f(x) = -k$

Where

$f(x) = x^4 + 4x^3 - 8x^2 = x^2(x^2 + 4x - 8)$

Let $g(x) = -k$



From the graph the following cases arise :

1. When $-3 \le -k \le 0, \Rightarrow 0 \le k \le 3$

In this case, $y = -k$ intersect at four points.

2. When $-4 \le -k < -3, \Rightarrow 3 < k \le 4$

In this case, $y = -k$ intersect at two points, the given equation has two real roots.

3. When $k < 0, \Rightarrow -k > 0$

In this case, there are two points of intersection. So, the equation has two real roots.

SECTION – C

49. 00030.00

Sol. $\frac{4000}{a_1 a_{4001}} = 10 \Rightarrow a_1 a_{4001} = 400$

also $a_1 + a_{4001} = 50$

$\Rightarrow |a_1 - a_{4001}|^2 = 2500 - 1600$

$\Rightarrow |a_1 - a_{4001}| = 30$

50. 00006.00

Sol. Let $z = x_1 + iy_1, w = x_2 + iy_2$

$\therefore |z - 3| = \text{Re}\, z \Rightarrow (x_1 - 3)^2 + y_1^2 = x_1^2$ ....(i)

$|w - 3| = \text{Re}\, w \Rightarrow (x_2 - 3)^2 + y_2^2 = x_2^2$ ...(ii)

$\arg(z - w) = \frac{\pi}{4} \Rightarrow \frac{y_1 - y_2}{x_1 - x_2} = \tan\frac{\pi}{4} = 1$ ...(iii)

By solving (i), (ii) and (iii) we get

$\Rightarrow \frac{y_1 - y_2}{x_1 - x_2} = \frac{6}{y_1 + y_2}$

From (iii) $y_1 + y_2 = 6$

51. 00028.00

Sol. If a_1 is mapped to 2, we have ${}^7C_5$ ways of mapping rest of the elements.
If a_1 is mapped to 3, we have ${}^6C_5$ ways of mapping rest of the elements.
If a_1 is mapped to 4, we have ${}^5C_5$ ways of mapping rest of the elements.
Hence total number of increasing functions = ${}^7C_5 + {}^6C_5 + {}^5C_5 = 28$.

52. 00007.00

Sol. $\left({}^{19}c_0(x^3)^{19} + {}^{19}c_1(px^2 + 2x - 5)(x^3)^{18} + ...\right)$

$\left({}^8c_0x^{16} + {}^8c_1(qx - 41)x^{14} + ...\right)\left({}^6c_0x^{24} + {}^6c_1(-x^3 + x - 1)x^{20} + ...\right)$

$= x^{97} + 391x^{96} + a_{95}x^{95} + ...$

Comparing the coefficient of x^{96} we get

$19p + 8q - 6 = 391$

$\Rightarrow 19p + 8q = 397$ Let $q = 19\lambda + k$, $0 \le k < 19$

$p = \frac{397 - 8(19\lambda + k)}{19} = 21 - 8\lambda - 2\frac{(4k+1)}{19}$

$\frac{4k+1}{19}$ must be integer $\Rightarrow k = 14$, $p = 15 - 8\lambda$

For minimum positive value of p, $\lambda = 1 \Rightarrow p = 7$

53. 00005.00

Sol. Let $\vec{d}.\vec{a} = \cos y\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] = -\vec{d}.\left(\vec{b}+\vec{c}\right)$ [as $\vec{d}.\left(\vec{a}+\vec{b}+\vec{c}\right)=0$]

$\Rightarrow \quad \cos y = -\frac{\vec{d}.\left(\vec{b}+\vec{c}\right)}{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]}$ .....(1)

similarly $\sin x = -\frac{\vec{d}.\left(\vec{b}+\vec{a}\right)}{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]}$ .....(2)

$\Rightarrow 2 = -\frac{\vec{d}.\left(\vec{a}+\vec{c}\right)}{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]}$ .....(3)

Adding these we get $\sin x + \cos y + 2 = 0$

$\Rightarrow \sin x + \cos y = -2$

$\Rightarrow \sin x = -1, \cos y = -1$

$\Rightarrow x = (4n-1)\frac{\pi}{2},\ y = (2n+1)\pi$

Since we want minimum value of $x^2 + y^2$, so $x = -\frac{\pi}{2},\ y = \pi$

$\Rightarrow x^2 + y^2 = \frac{5\pi^2}{4} \Rightarrow \lambda = 5$

54. 00002.00

Sol. Ans. (2)

Put $x = t^2$